

SECTION 1: ON-DEVICE CLIENT PIPELINE

Single Shared Backbone feeds Tripartite Heads with Active Masking



Private Edge Dataset

- Extreme non-IID label skew across heterogeneous nodes
- *Raw samples remain strictly isolated on-device at all times*



Single Shared Feature Backbone

- Unified feature extractor across all classification heads
- **Completely eliminates duplicate network instances**
- *Minimal edge footprint: 50% lower VRAM, fast inference*



Active-Class Logit Masking (ACLM Gate)

- Masks unobserved class logits before loss evaluation
- **Blocks negative gradient interference under label skew**
- *Preserves shared geometric feature representations*



Global Root Head

*Global Consensus
Synchronized with
Federated Server*

Broad Generalization

Parent Head

*Peer Cluster Cohort
Synchronized with
Peer Cluster*

Synergistic Support

Private Local Head

*100% On-Device
Never Uploaded
Zero Telemetry*



Calibrated Ensemble Inference

*Anchored linear combination of temperature-calibrated logits
Optimal personalization while safeguarding tail decile fairness*

SECTION 2: MULTI-TIER FEDERATION TOPOLOGY

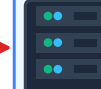
Hierarchical Synchronization • Subspace Defense • Affinity Peer Collaboration

Central Federated Server (Global Consensus)



Subspace Byzantine Defense

- Subspace Cosine Filtering (SCCF)
- Temporal Trust Tracking (TTT)
- Neutralizes poisoning attacks*



Consensus Aggregator

- Shared Backbone Updating
- Global Root Head Averaging
- Trust-weighted parameter fusion*

Defended Updates

Consensus Broadcast

Collaborative Peer Clusters (Cohort Layer)

Dynamic grouping via 256-dim Random Projection sketches • Synchronizes Parent Heads

Peer Cluster 1

(Cohort A)

Parent Head Model

Synergistic Peer Fusion

Peer Cluster 2

(Cohort B)

Parent Head Model

Synergistic Peer Fusion

Peer Cluster K

(Cohort K)

Parent Head Model

Synergistic Peer Fusion



Heterogeneous Edge Client Fleet

Client 1

Mobile Device

Cluster 1 Member



Client 2

Edge Server

Cluster 2 Member



Client K

IoT Node

Cluster K Member



— Global Consensus Sync (Root Head & Backbone)

- - - Peer Cluster Sync (Parent Head)



100% On-Device Isolation